

# **BSc (Hons) in Computer Systems Engineering**

**Semester 1, 2025**

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## **Practical Assignment**

**Design, Construction, and Testing of a Multi-Stage Discrete Audio Output  
Circuit for AM Broadcast Receive**

**IE2034 – Analog Electronics**

**Group Number – 08 :**

**IT23657946 - PERERA M S L**

**IT23658486 - DISSANAYAKE D M J T**

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## 1. Objective

The objective of this experiment is to design, build, and test the audio output circuit of a basic AM broadcast receiver using discrete components (BJTs, diodes, resistors, and capacitors). The circuit should:

1. Filter the detected AM signal to isolate the audio frequency range of 1–20 kHz using a band-pass filter.
2. Amplify the weak audio signal using a common-emitter BJT amplifier with a gain of approximately 20.
3. Deliver a clean and identifiable audio output, which can be observed visually or audibly.
4. Demonstrate the theoretical calculations, design choices, and practical performance of each stage.

## 2. Introduction

Amplitude Modulation (AM) is a technique used in radio broadcasting where the amplitude of the carrier wave is varied according to the audio signal. An AM receiver detects these signals and extracts the original audio information.

The detected audio signal from an AM receiver is usually weak and contains unwanted frequency components. Therefore, it requires:

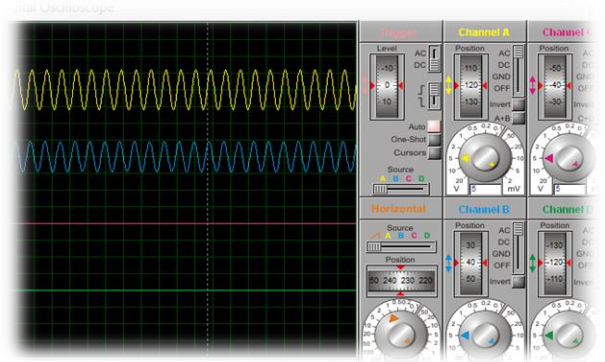
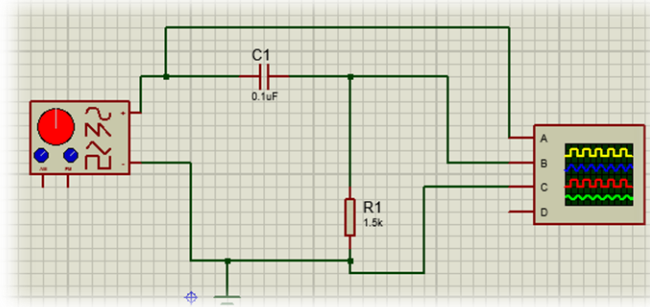
- **Filtering:** To remove noise and restrict the signal to the audible range (1–20 kHz). A band-pass filter combining high-pass and low-pass filters is used for this purpose.
- **Amplification:** To increase the signal amplitude to a level suitable for driving an output device such as a speaker or LED indicator. A common-emitter BJT amplifier is used because it provides significant voltage gain and inverts the signal.

This project focuses on constructing a multi-stage discrete circuit without ICs or transformers, ensuring understanding of fundamental analog electronics, including filter design, BJT biasing, and signal amplification.

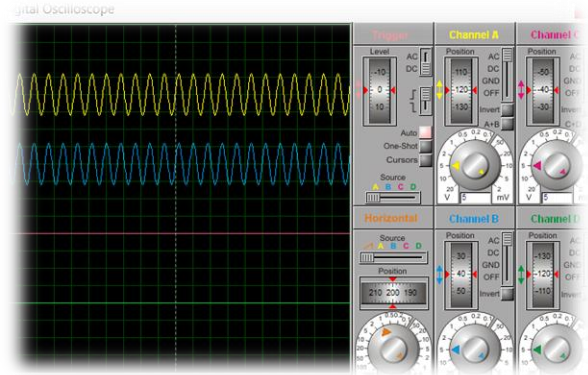
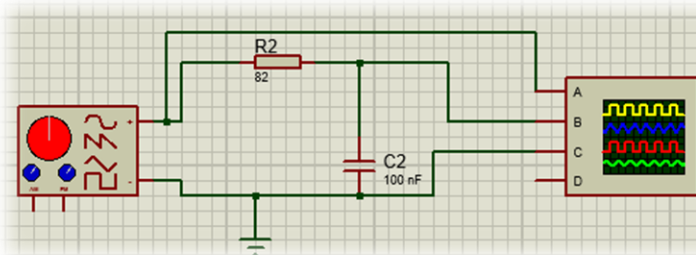
### 3. Design Overview

#### 3.1 Block Diagram

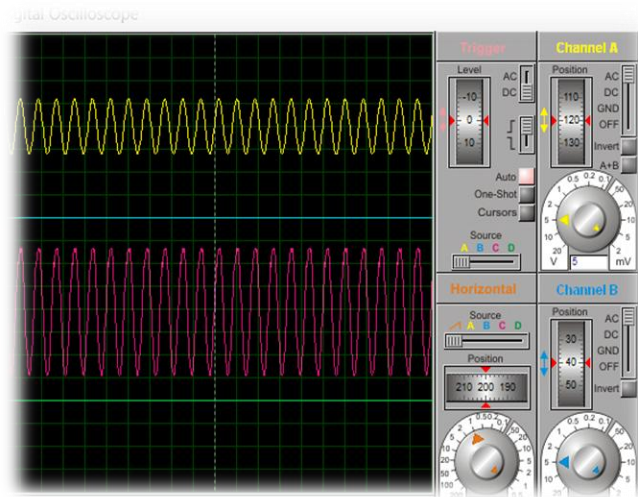
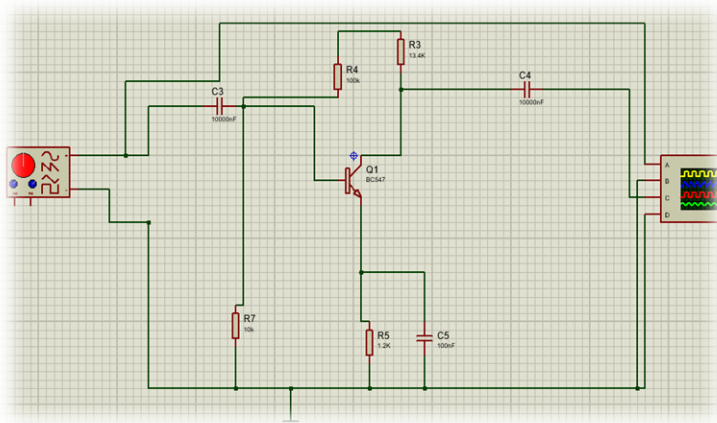
##### 3.1.1 High pass filter



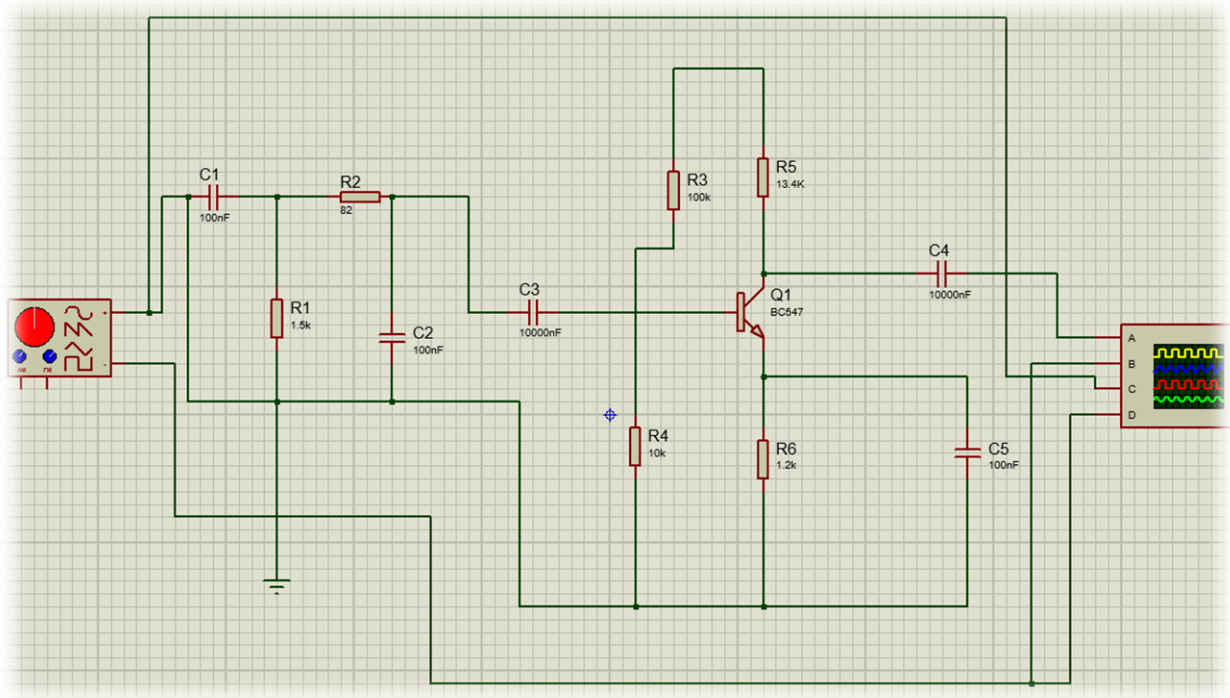
##### 3.1.2 Low pass filter



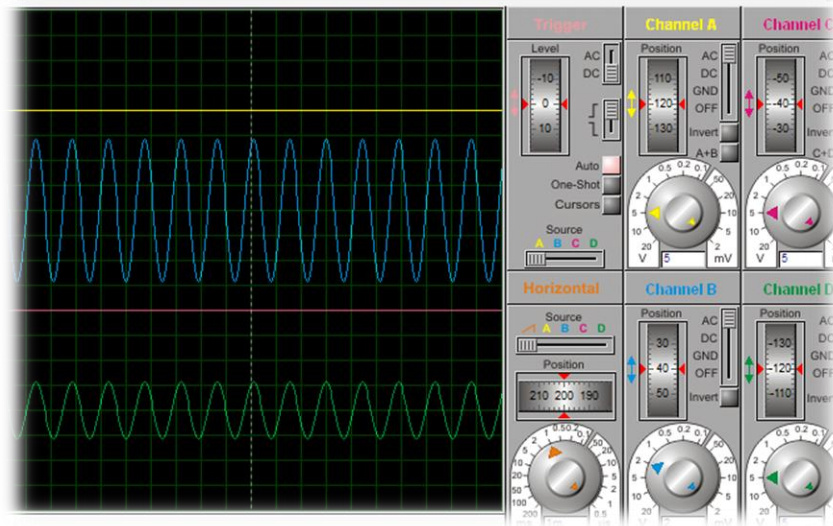
##### 3.1.3 BJT Amplifier



### 3.1.4 Overall Block diagram



Digital Oscilloscope



## 4. Design of Components

### Frequency Range (1–20 kHz):

The circuit is designed to process audio signals within the frequency range of 1 kHz to 20 kHz. A band-pass filter is used to allow only this range of frequencies to pass through while attenuating unwanted low- and high-frequency noise. The high-pass section removes frequencies below 1 kHz, and the low-pass section limits frequencies above 20 kHz, ensuring clear and noise-free audio output.

### Choice of Components (BJT, Resistors, Capacitors):

Discrete components are used to better understand analog behavior without relying on integrated circuits.

- **BJT:** A general-purpose NPN transistor (e.g., BC547 or 2N2222) is used as a common-emitter amplifier to achieve a voltage gain of about 20.
- **Resistors:** Used for biasing, load, and stabilization in the amplifier, and also in the RC filter networks to set cutoff frequencies.
- **Capacitors:** Employed in coupling, bypassing, and filter stages. The RC combinations are selected using  $f_c = 1/2\pi RC$  to achieve desired cutoff points at 1 kHz (high-pass) and 20 kHz (low-pass).

### Stability and Biasing:

A proper DC biasing network is provided for the BJT to ensure operation in the active region, preventing distortion or cutoff in the amplified signal. The bias resistors are chosen to maintain a stable base voltage and collector current. A bypass capacitor across the emitter resistor enhances AC gain while maintaining DC stability. Decoupling capacitors are also used to minimize noise and stabilize the power supply.

## 5. High-pass Filter

### 5.1 Purpose

- The purpose of the high-pass filter is to remove unwanted low-frequency signals and DC components from the input. It allows only the audio frequencies above 1 kHz to pass through to the next stage. This helps eliminate noise such as hum or slow variations, ensuring that only useful audio signals reach the amplifier.

### 5.2 Theoretical Calculations

- Cutoff Frequency Calculation:

$$F_c = 1/2\pi RC$$

$$1*10^3 = 1/(2\pi * R * 0.1*10^{-6})$$

$$R = 1/2\pi 10^3 * 0.1*10^{-6}$$

$$R = 1591.5\Omega$$

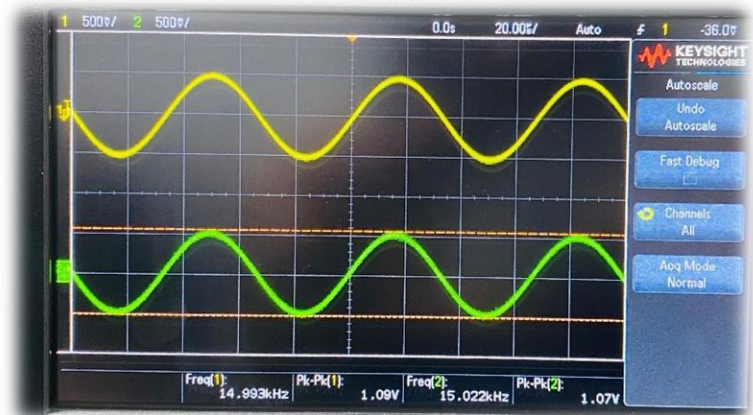
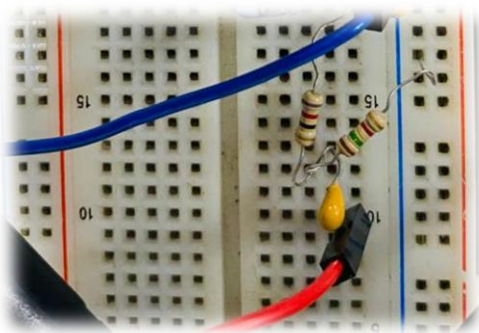
$$R = 1.591K\Omega$$

(Practical value obtained: 1.5K)

### 5.3 Component Selection

- Resistor (R): 1.6 k $\Omega$  — chosen based on the cutoff frequency formula  $f_c = 1/2\pi RC$  for 1 kHz.
- Capacitor (C): 0.1  $\mu$ F (non-polarized) — gives stable performance for audio frequencies.
- Reason: These values ensure only signals above 1 kHz pass through while blocking DC and low-frequency noise.

### 5.4 Circuit Diagram



### 5.5 Observation

- When the input signal was applied to the high-pass filter, the oscilloscope showed that low-frequency signals (below 1 kHz) were greatly reduced, while signals above 1 kHz passed through clearly. The output waveform had smaller amplitude at low frequencies and increased as the frequency went higher. This confirms that the circuit removed unwanted low-frequency noise and DC components, allowing only the desired audio frequencies to pass to the amplifier stage.

## 6. Low-pass Filter

### 6.1 Purpose

- The purpose of the low-pass filter is to remove high-frequency noise and allow only the useful audio signals (below 20 kHz) to pass. This ensures the output sound is smooth and free from unwanted high-frequency distortion before amplification or output.

### 6.2 Theoretical Calculations

- Cutoff Frequency Calculation:  

$$F_c = 1/2\pi RC$$

$$20 \times 10^3 = 1/(2\pi \times R \times 0.1 \times 10^{-6})$$

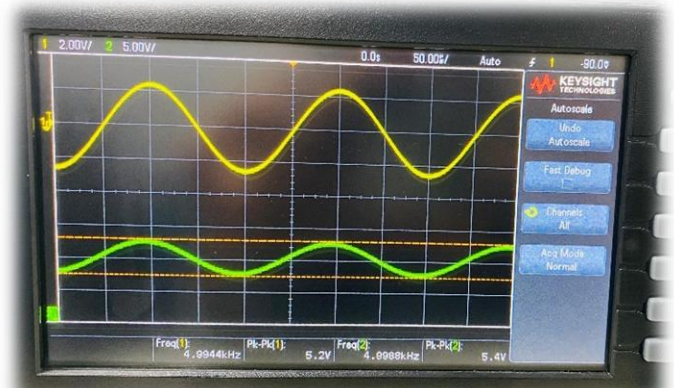
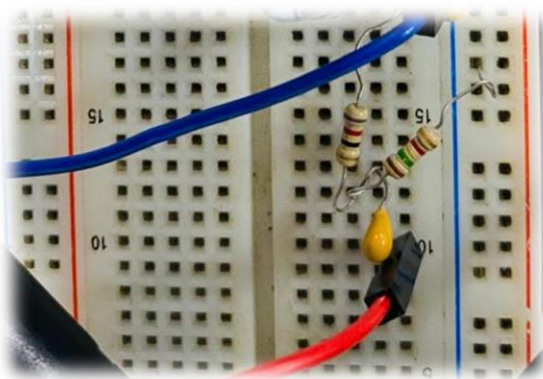
$$R = 1/2\pi \times 20 \times 10^3 \times 0.1 \times 10^{-6}$$

$$R = 79.6\Omega$$
 (Practical value obtained: 82Ω)

### 6.3 Component Selection

- Resistor (R): 82 Ω, metal-film type for low noise and stability.
- Capacitor (C): 0.1 μF (100 nF), non-polarized polyester or ceramic capacitor for stable audio performance.
- These values give a cutoff frequency close to 20 kHz, which passes audio frequencies below this limit and removes unwanted high-frequency noise.
- Components are connected in a simple RC network: resistor in series, capacitor to ground.

### 6.4 Circuit Diagram



### 6.5 Observation

- When the input signal was applied, the oscilloscope showed that high-frequency components above 20 kHz were attenuated, while lower frequencies passed with minimal loss. The output waveform appeared smooth and clean compared to the input. This confirms that the low-pass filter effectively removed high-frequency noise, providing a stable audio signal for the next circuit stage.

## 7. BJT Amplifier

### 7.1 Purpose

- The purpose of the BJT amplifier is to increase the amplitude of weak input signals using a bipolar junction transistor. It converts a small input voltage or current into a larger output voltage or current while maintaining the same waveform shape.

### 7.2 Theoretical Calculations

Given data:

In this design, the values of  $R_2$  and  $R_C$  were not given, so we assumed  $R_2=10k\Omega$  and  $R_C=9k\Omega$  for the calculations.

$V_{CC}=12V$ ,  $V_C=6V$ ,  $R_C=9k\Omega$ ,  $R_2=10k\Omega$ , Desired voltage gain  $AV=20A$

#### 1) Collector Current $I_C$

$$\text{So, } V_C = I_C \times R_C$$

$$I_C = \frac{V_C}{R_C}$$

$$I_C = \frac{6V}{9K} = 0.66mA$$

$$\text{Since } I_E \approx I_C: \quad I_E = 0.66mA$$

#### 2) Emitter Resistance

$$AV = \frac{R_C}{R_E}$$

$$R_E = \frac{R_C}{AV} = \frac{9K}{20} = 0.5K\Omega$$

(Practical value obtained:  $1.2k\Omega$ )

#### 3) Small Signal Emitter Resistance

Terminal Voltage =  $25mA$

$$r_e = \frac{25 \text{ mV}}{I_E} = \frac{25 \times 10^{-3}}{0.66 \times 10^{-3}} = 38 \Omega$$

#### 4) With load resistance

$R_C = 9 k\Omega$  and external load  $R_L = 1 k\Omega$ .

Parallel combination:

$$R_{C||L} = \frac{R_C \times R_L}{R_C + R_L} = \frac{90000 \times 1000}{90000 + 1000}$$

So,

$$R_{C||L} = \frac{90\,000\,000}{91000} = 989.01\, \Omega = 989\, \Omega$$

Voltage gain  $A_v$  (with emitter bypassed)

$$A_v = -\frac{R_{C||L}}{r_e} = -\frac{989}{38} = -26.02$$

Hence, Voltage Gain  $\approx 26.01$

#### 5) Base Bias Calculations

Base voltage:

$$V_E = I_E \times R_E$$

$$= 0.66\text{mA} \times 0.5\, \text{k}\Omega$$

$$= 0.33\text{V}$$

$$V_{BE} = V_B - V_E$$

$$V_B = V_{BE} + V_E$$

$$V_B = 0.7 + V_E$$

$$= 0.7 + 0.33 = 1.03\, \text{V}$$

$$V_B = V_{CC} \times \left( \frac{R_2}{R_1 + R_2} \right)$$

$$\left( \frac{R_2}{R_1 + R_2} \right) = \frac{V_B}{V_{CC}}$$

$$= \frac{1.03\text{V}}{12\text{V}} = 0.085$$

Assume  $R_2 = 10\text{k}\Omega$ :

$$\left( \frac{R_2}{R_1 + R_2} \right) = 0.085$$

$$= \left( \frac{10\text{k}}{R_1 + 10\text{k}} \right) = 0.085$$

$$= R_1 = 107.6\text{k}\Omega \text{ (Practical value obtained: } 100\text{k}\Omega)$$

6) Assume  $\beta = 100$

Base current :  $I_B = \frac{I_C}{\beta} = \frac{0.66 \text{ mA}}{100} = 6.6 \mu\text{A}$

7) Input coupling capacitor  $C_{in}$

compute the bias divider parallel (Thevenin resistance)

$R_1 = 100 \text{ k}\Omega$ ,  $R_2 = 10 \text{ k}\Omega$

$$R_{th} = R_1 \parallel R_2 = \frac{100000 \times 10000}{100000 + 10000}$$

$$R_{th} = 9.091 \text{ K}\Omega$$

transistor input impedance.

$$(\beta + 1) r_e = 101 \times 38 = 3.83 \text{ K}\Omega$$

effective input resistance  $R_{in}$

Now  $R_{in} = R_{th} \parallel (\beta + 1) r_e$ .

$$R_{in} = \frac{9,091 \times 3838}{9,091 + 3838}$$

$$R_{in} = 2.69 \text{ K}\Omega$$

We want the cutoff well below the 1 kHz passband edge. So, Use  $f_{c,in} = 20 \text{ Hz}$  as a conservative design target, we want the low cutoff  $\leq 20 \text{ Hz}$  (so that 1–20 kHz audio passes clearly).

$$C_{in} = \frac{1}{2\pi f_{c,in} R_{in}}$$

$$C_{in} = \frac{1}{2\pi \times 20 \times 2698} = 2.34 \mu\text{F}$$

Choose nearest standard value  $\rightarrow 10 \mu\text{F}$

8) Output Coupling Capacitor ( $C_{out}$ )

Choose  $f_{L2} = 20 \text{ Hz}$ ,  $R_{Load} = 1 \text{ k}\Omega$ :

$$C_{out} = \frac{1}{2\pi(1000)(20)} = \frac{1}{125660} = 7.96 \mu\text{F}$$

Choose standard value  $\rightarrow 10 \mu\text{F}$

9) Emitter Bypass Capacitor ( $C_E$ )

Bypasses the AC signal around the emitter resistor to increase gain.

$$f_{LE} = \frac{1}{2\pi R_E C_E}$$

Rearrange for  $C_E$ :

$$C_E = \frac{1}{2\pi R_E f_{LE}} = \frac{1}{2\pi(1200)(20)} = \frac{1}{150,796.44} = 6.3\mu F$$

Choose the next higher standard value 0.1 $\mu$ F

#### 10) Q-Point Calculation

- Base voltage (VB)

$$V_B = V_{CC} \times \frac{R_2}{R_1 + R_2} = 12 \times \frac{10}{110} = 1.09V$$

- Emitter voltage (VE)

$$V_E = V_B - 0.7 = 1.03 - 0.7 = 0.33V$$

- Emitter current (IE)

$$I_E = \frac{V_E}{R_E} = \frac{0.39}{500} = 0.7 \text{ mA}$$

- Collector current (IC) =  $I_E = 0.3\text{mA}$

- Collector voltage (VC)

$$V_C = V_{CC} - (I_C \times R_C) = 12 - (0.7\text{mA} \times 9\text{k}\Omega) = 12 - 6.3V = 5.7V$$

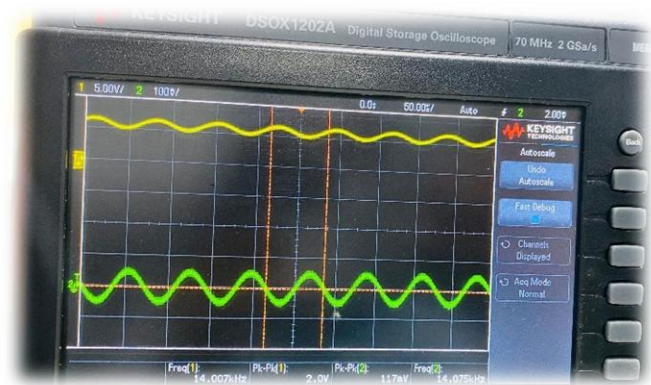
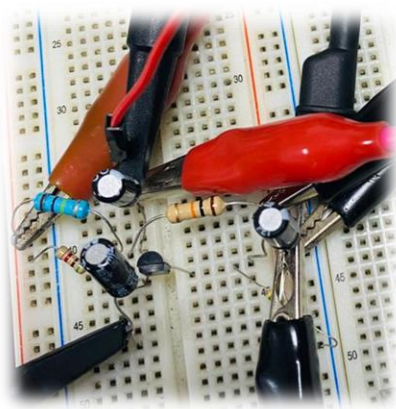
#### 11) Comparison of Theoretical and Practical Values

| Component / Parameter         | Theoretical Values      | Practical Value |
|-------------------------------|-------------------------|-----------------|
| High-pass resistor (R)        | 1.591 k $\Omega$        | 1.5 k $\Omega$  |
| Low-pass resistor (R)         | 76 $\Omega$             | 82 $\Omega$     |
| Collector resistor (RC)       | 9 k $\Omega$            | 13.4 k $\Omega$ |
| Emitter resistor (RE)         | 0.5 k $\Omega$          | 1.2 k $\Omega$  |
| Bias resistor R2              | 10 k $\Omega$ (Assumed) | 10 k $\Omega$   |
| Bias resistor R2              | 107.6 k $\Omega$        | 100k $\Omega$   |
| Input capacitor (Cin)         | 2.4 $\mu$ F             | 10 $\mu$ F      |
| Output capacitor (Cout)       | 7.96 $\mu$ F            | 10 $\mu$ F      |
| Emitter bypass capacitor (CE) | 6.3 $\mu$ F             | 0.1 $\mu$ F     |
| Voltage gain                  | 26.02V                  | 30.4V           |
| Q-point                       | 5.7V                    | 6.2V            |

### 7.3 Component Selection

- Transistor (BJT): Acts as the main amplifying device (e.g., BC547 or 2N3904) due to its good gain and frequency response.
- Resistors ( $R_1$ ,  $R_2$ ,  $R_C$ ,  $R_E$ ): Used to bias and stabilize the transistor's operating point (Q-point).
- Capacitors ( $C_{in}$ ,  $C_{out}$ ,  $C_E$ ):
  - $C_{in}$ : Couples the input signal to the base.
  - $C_{out}$ : Couples the amplified output signal to the load.
  - $C_E$ : Provides AC bypass for the emitter resistor to increase gain.
- Power Supply ( $V_{CC}$ ): Provides the required DC bias voltage (typically 12V)

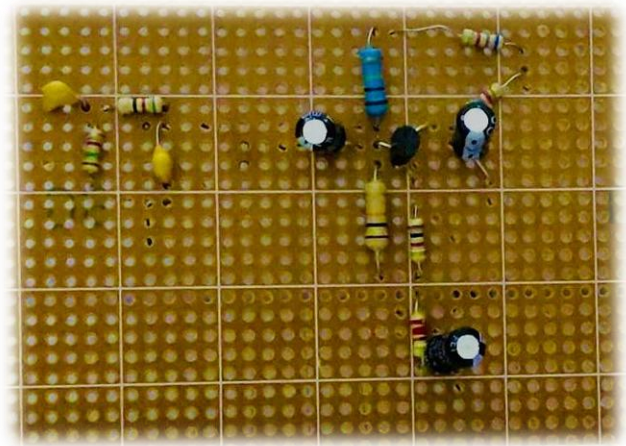
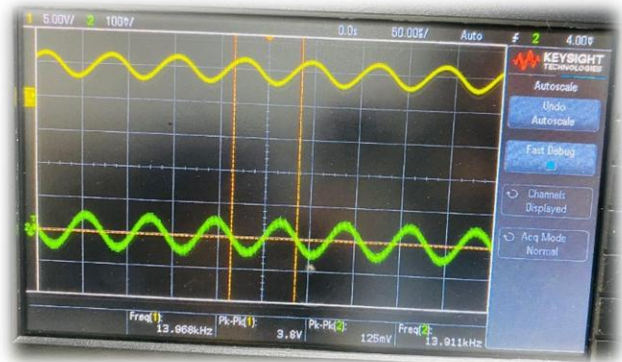
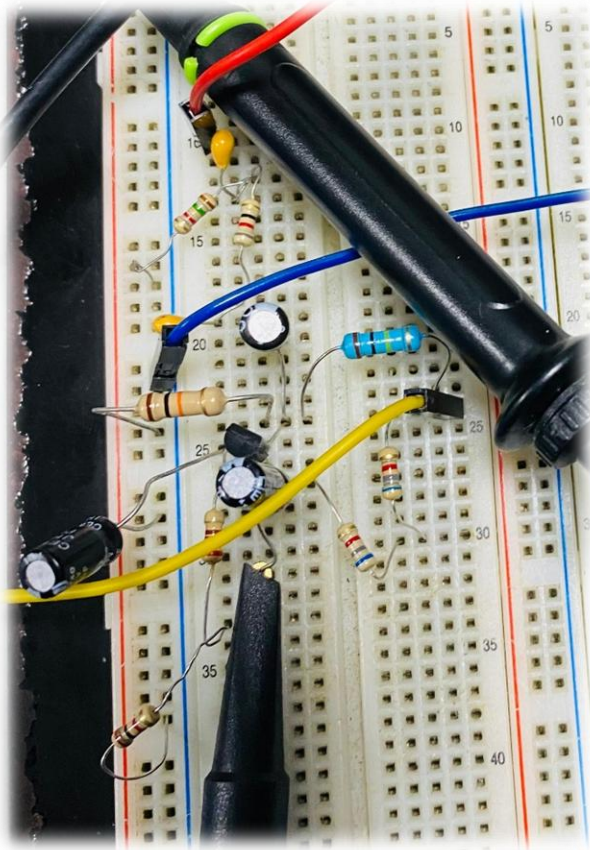
### 7.4 Circuit Diagram



#### a. Observation

- When the input signal was applied to the BJT amplifier circuit, the output waveform showed a clear increase in amplitude, confirming signal amplification. The output signal maintained the same waveform shape as the input but with higher voltage. The oscilloscope readings indicated that the amplifier provided a voltage gain of about 23.6, close to the expected theoretical value of 20.

## 8. Complete Circuit



- We measured the Q-point as 6.21 V.



- Gain (from oscilloscope reading) =  $3.8/0.125$   
Therefore, the amplifier voltage gain is 30.4

## 9. Experimental Procedure

- Constructed the high-pass filter, amplifier, and low-pass filter stages separately on the dot board.
- Verified the operation of each stage individually using an oscilloscope before combining them.
- For the high-pass filter, applied input signals from a function generator and observed the cutoff frequency around 1 kHz.
- For the BJT amplifier, adjusted the biasing resistors to ensure the transistor operated in the active region and measured a voltage gain of approximately 20.
- For the low-pass filter, tested input signals up to 20 kHz and observed signal attenuation beyond this frequency.

## 10. Results

- The high-pass filter successfully removed low-frequency noise and DC components below 1 kHz.
- The amplifier provided a voltage gain of approximately 30.4, amplifying weak signals to a clearly visible waveform.
- The low-pass filter effectively reduced high-frequency noise above 20 kHz.
- The combined circuit provided a clean, audible audio output with minimal distortion.
- The oscilloscope readings matched well with calculated cutoff frequencies, and the measured amplifier gain confirmed the expected signal amplification.

## 11. Discussion

The experiment demonstrated how a multi-stage circuit can be used to process and recover audio signals from a weak RF detector output. The high-pass and low-pass filters defined the desired audio band (1–20 kHz), while the BJT amplifier increased signal strength for clear output. Small deviations between theoretical and measured values occurred due to resistor tolerances, capacitor variations, and wiring effects on the dot board. Overall, the design achieved the expected performance, showing proper filtering and stable amplification.

## 12. Conclusion

The designed audio output circuit successfully filtered and amplified the input signal within the 1–20 kHz range. Each stage operated as intended: the high-pass filter removed unwanted low frequencies, the amplifier boosted the signal with stable gain, and the low-pass filter eliminated high-frequency noise. The final output was clean and strong enough to drive an indicator or speaker, meeting the design objectives.

## 13. References

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